UNIT 6 · FUNCTIONS · 8TH GRADE MATH

Tables & Systems of Equations

Name _____ Date _____ Period _____

Guided Notes

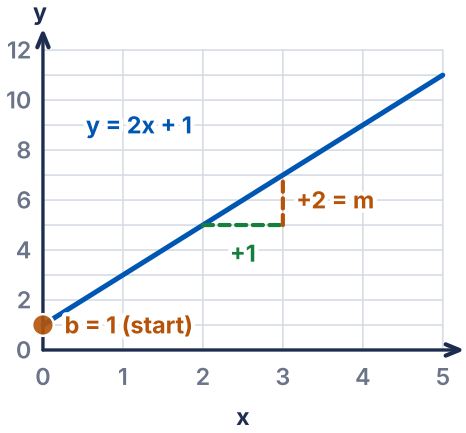
1 Writing Linear Equations from a Table

The Big Idea

Every linear equation can be written as **y = mx + b**. To go from a table to an equation, we need to find two things:

- **m** = the slope (the rate of change)
- **b** = the y-intercept (the y-value when **x = 0**)

$$y = mx + b$$



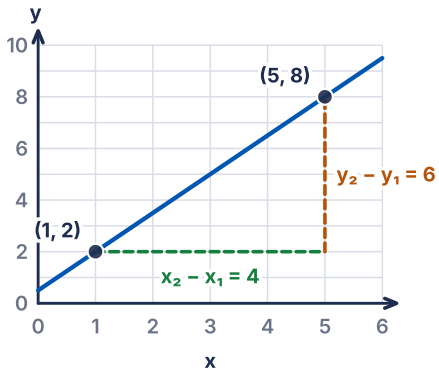
b = where the line starts on the y-axis. **m** = how much y changes each time x goes up 1.

> **QUICK CHECK: IS THE TABLE LINEAR?**
> A table shows a linear relationship only when the y-values change by the *same amount* for equal steps in x. If x goes up by 1 each row and y jumps around, it's not linear.

Finding the Slope from a Table

$$m = (y_2 - y_1) \div (x_2 - x_1) = \text{change in } y \div \text{change in } x$$

Pick any two rows of the table. Subtract to get the change in y and the change in x. Divide.



m = 6 ÷ 4 = 1.5

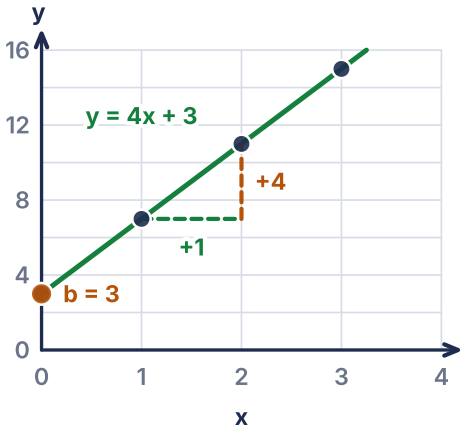
EXAMPLE 1 **Table includes x = 0**

x	y
0	3
1	7
2	11
3	15

1. **Find the slope.** From row 1 to row 2: y changes by `+4`, x changes by `+1`. So `m = 4`.
2. **Find the y-intercept.** When x = 0, y = 3. So **`b = 3`**.
3. **Write the equation:** `y = 4x + 3`

x	y	Δy ÷ Δx
0	3	
1	7	$\frac{4}{1} = 4$
2	11	$\frac{4}{1} = 4$
3	15	$\frac{4}{1} = 4$

+1, +4 between each pair of rows

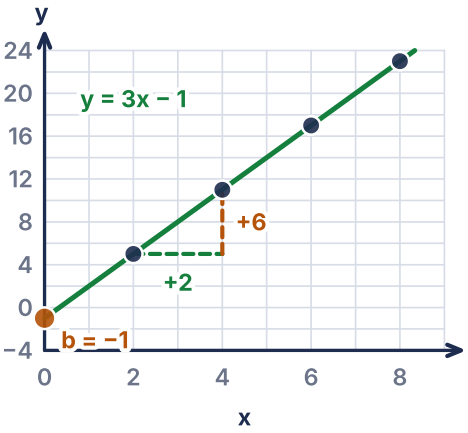


EXAMPLE 2 **Table does NOT include x = 0**

x	y
2	5
4	11
6	17
8	23

1. **Find the slope.** From (2, 5) to (4, 11): Δy = 6, Δx = 2, so $m = \frac{6}{2} = 3$.
2. **Work backward to find b.** Start at `(2, 5)`. Back up x by 1 means y drops by 3. So at `x = 1, y = 2`. At `x = 0, y = −1`. So `b = −1`.
3. **Write the equation:** `y = 3x − 1`
4. **Check.** Plug in x = 8: y = 3(8) − 1 = 23. ✓ Matches the table.

x	y	Δy ÷ Δx
0	−1	
		$\frac{3}{1} = 3$
1	2	
		$\frac{3}{1} = 3$
2	5	
		$\frac{6}{2} = 3$
4	11	
		$\frac{6}{2} = 3$
6	17	
		$\frac{6}{2} = 3$
8	23	

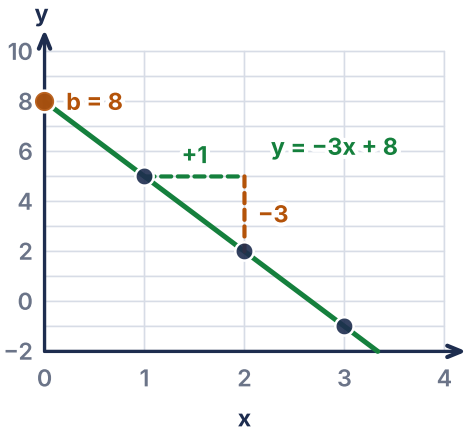


EXAMPLE 3 **Negative slope (y decreases as x increases)**

x	y
0	8
1	5
2	2
3	−1

1. **Find the slope.** From (0, 8) to (1, 5): Δy = `−3`, Δx = `+1`. So `m = −3`. (When y *decreases* as x increases, the slope is negative.)
2. **Find the y-intercept.** When x = 0, y = 8. So `b = 8`.
3. **Write the equation:** `y = −3x + 8`
4. **Check.** Plug in x = 3: y = −3(3) + 8 = −1. ✓ Matches the table.

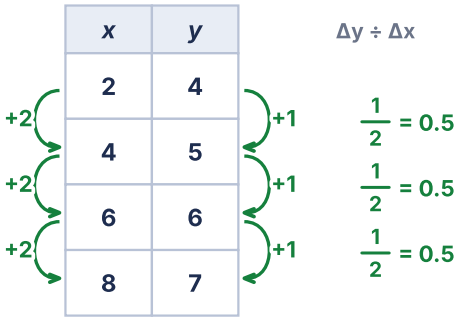
x	y	Δy ÷ Δx
0	8	
		$\frac{-3}{1} = -3$
1	5	
		$\frac{-3}{1} = -3$
2	2	
		$\frac{-3}{1} = -3$
3	−1	

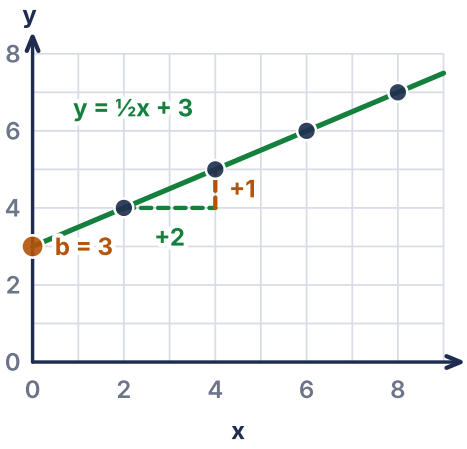


EXAMPLE 4 **Fractional slope**

x	y
2	4
4	5
6	6
8	7

1. **Find the slope.** From (2, 4) to (4, 5): Δy = `1`, Δx = `2`. So `m =` $\frac{1}{2}$. (Don't round — keep it as a fraction.)
2. **Work backward to find b.** Start at `(2, 4)`. Back up x by 2 (to x = 0) means y drops by 1. So at `x = 0, y = 3`. So `b = 3`.
3. **Write the equation:** `y =` $(\frac{1}{2})$`x + 3`
4. **Check.** Plug in x = 8: y = $(\frac{1}{2})(8) + 3 = 4 + 3 = 7$. ✓ Matches the table.





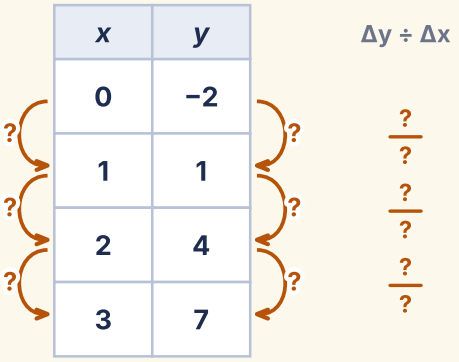
CHALLENGE: A SHORTCUT FOR FINDING B

Once you know *m*, pick **any** row (x, y) from the table and solve: `b = y − m·x`

Example 2 check: `b = 5 − 3(2) = −1`. ✓

▶ Try It · Table A

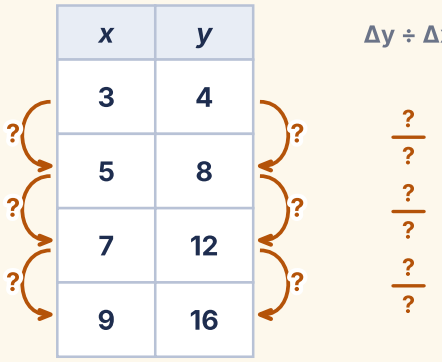
x	y
0	−2
1	1
2	4
3	7



m = _____ **b =** _____

▶ Try It · Table B (no x = 0)

x	y
3	4
5	8
7	12
9	16



No x = 0 row? After you find m, work backward up the table to x = 0.

m = _____ **b =** _____

✏ Practice: Writing Equations from a Table

1. ★ Try it What is the slope **m** of this table? Write just the number.

x	y
0	5
1	7
2	9
3	11

Answer: _____

2. ★★ On level Write the equation for this table in the form y = mx + b.

x	y
0	10
1	6
2	2
3	−2

Answer: _____

3. ★★★ Challenge This table has no x = 0 row. Find **b**, the y-intercept. Write just the number.

x	y
2	1
4	7
6	13
8	19

Answer: _____

2 Solving Systems by Graphing

What Is a System of Equations?

A **system of equations** is two or more equations that share the same variables. In 8th grade, we work with systems of **two linear equations** in two variables (x and y).

> EQ 1 $y = 2x + 1$
>
> EQ 2 $y = -x + 4$

x	y
0	1
1	3
2	5
3	7

EQ 1: y = 2x + 1

x	y
0	4
1	3
2	2
3	1

EQ 2: y = −x + 4

Both tables have the row x = 1, y = 3. That shared pair works in **both** equations.

Graphing Method: The Big Idea

Graph each line on the same coordinate plane. The point where the two lines **intersect** is the solution.

```
solution of a system = intersection point (x, y)
```

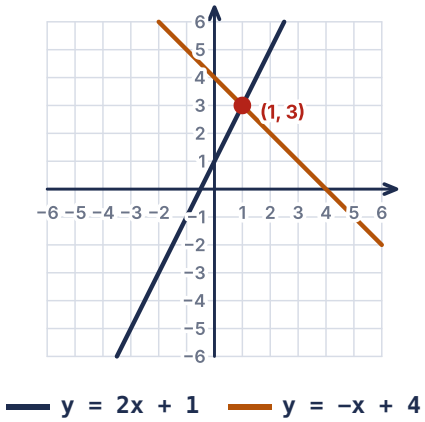
STEPS

1. Graph the first line (use slope and y-intercept, or a table).
2. Graph the second line on the same plane.
3. Find where they cross. That ordered pair is the solution.
4. Check by plugging the point into both equations.

EXAMPLE **Solve by graphing**

EQ 1 $y = 2x + 1$

EQ 2 $y = -x + 4$



1. **EQ 1:** slope 2, y-int 1. Graph from (0, 1): right 1, up 2 → (1, 3), (2, 5)...
2. **EQ 2:** slope −1, y-int 4. Graph from (0, 4): right 1, down 1 → (1, 3), (2, 2)...
3. **Read the intersection:** `(1, 3)`
4. **Check.** EQ 1: 3 = 2(1) + 1 = 3 ✓. EQ 2: 3 = −(1) + 4 = 3 ✓. Both true. **Solution: (1, 3)**.

Three Possible Outcomes

When you graph a system of two lines, one of three things will happen:

CASE 1 · ONE SOLUTION

The two lines **cross at exactly one point**. The system has **one solution** — that point. (Different slopes.)

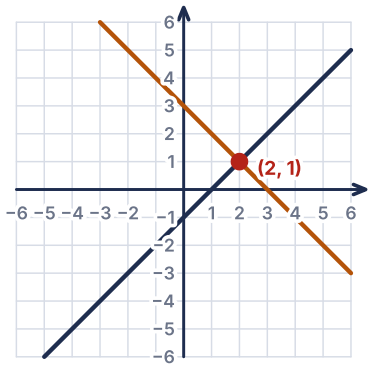
CASE 2 · NO SOLUTION

The two lines are **parallel**. They never cross. The system has **no solution**. (Same slope, different y-intercepts.)

CASE 3 · INFINITELY MANY SOLUTIONS

The two equations are the **same line**. Every point on the line works. **Infinitely many solutions**. (Same slope AND same y-intercept.)

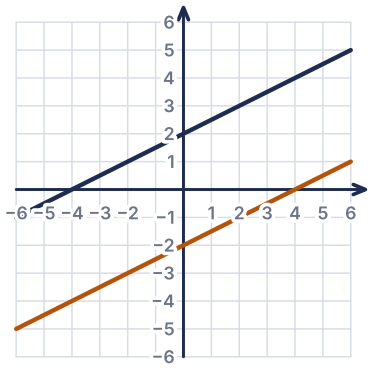
CASE 1 · ONE SOLUTION



They cross once

$y = x - 1$ $y = -x + 3$

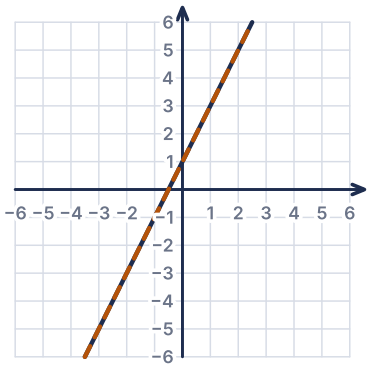
CASE 2 · NO SOLUTION



Parallel — never cross

$y = \frac{1}{2}x + 2$ $y = \frac{1}{2}x - 2$

CASE 3 · INFINITELY MANY



One line, drawn twice

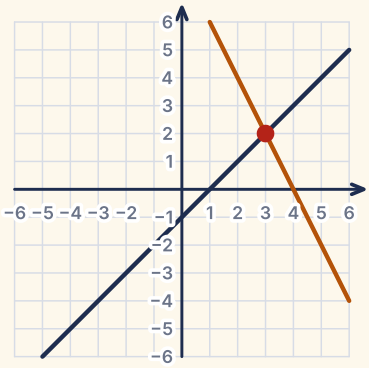
$y = 2x + 1$ $y = 2x + 1$

Count the intersection points: **one** point → one solution, **zero** points → no solution, **every** point → infinitely many.

▶ Try It · Read the Graph

Read the graph: the two lines cross at the red point. Write the solution as an ordered pair.

Solution: (_____ , _____)

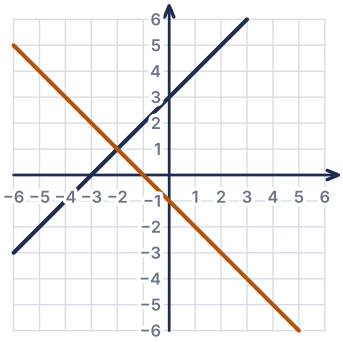


$y = x - 1$ $y = -2x + 8$

✏ Practice: Solving Systems by Graphing

1. ★ Try it Where do the two lines cross? Write the ordered pair, like (2, 3).

Answer: _____

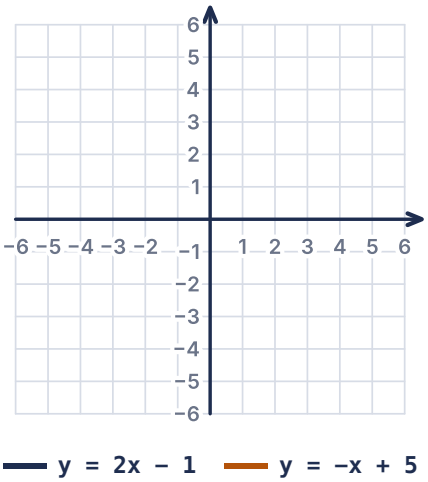


$y = x + 3$

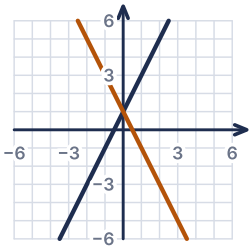
$y = -x - 1$

2. ★★ On level Graph both lines on the grid. Where do they cross? Write the ordered pair.

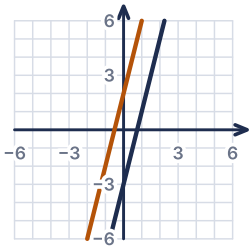
Answer: _____



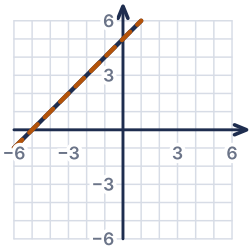
3. ★★★ Challenge Which system has **no solution**?



y = 2x + 1 and
y = −2x + 1



y = 4x − 3 and
y = 4x + 2



y = x + 5 and
y = x + 5

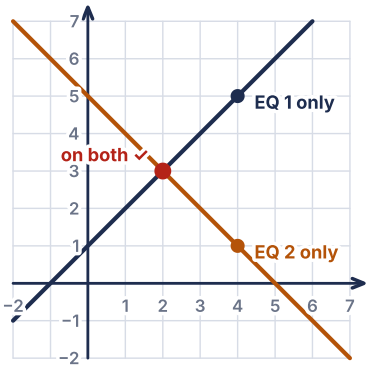
- ○ y = 2x + 1 and y = −2x + 1
- ○ y = 4x − 3 and y = 4x + 2
- ○ y = x + 5 and y = x + 5

3 What Does "Solution" Actually Mean?

The Core Definition

A **solution** to a system of equations is an ordered pair (x, y) that makes **every** equation in the system true at the same time.

> **TWO WAYS TO SAY IT**
>
> - **Graphically**: the solution is the point where the lines cross.
> - **Algebraically**: the solution is the (x, y) that makes both equations true.



y = x + 1 **y = −x + 5**

Only the crossing point (2, 3) sits on **both** lines, so only it is a solution.

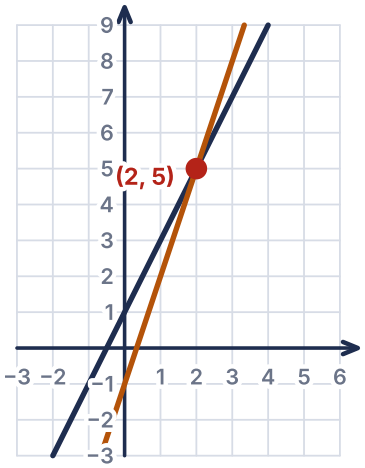
Testing Whether a Point Is a Solution

To test a candidate point (a, b), plug **x = a** and **y = b** into **each** equation. If the point makes *both* equations true, it's a solution. If it fails even one, it's **not** a solution.

EXAMPLE **Is (2, 5) a solution?**

> **EQ 1** **y = 2x + 1**
>
> **EQ 2** **y = 3x − 1**

1. **Test EQ 1** with x = 2, y = 5:
   `5 = 2(2) + 1 = 5` ✓ True.
2. **Test EQ 2** with x = 2, y = 5:
   `5 = 3(2) − 1 = 5` ✓ True.
3. Both equations are true, so **(2, 5) is a solution**.



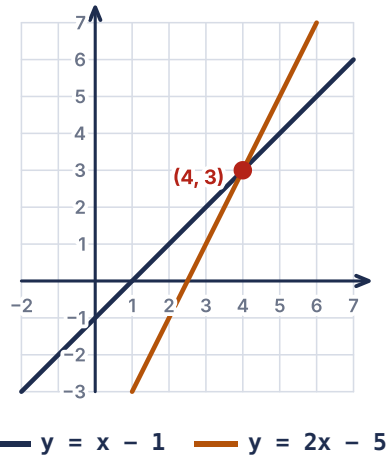
y = 2x + 1 **y = 3x − 1**

EXAMPLE **Is (4, 3) a solution?**

EQ 1 $y = x - 1$

EQ 2 $y = 2x - 5$

1. **Test EQ 1** with x = 4, y = 3:
   $3 = 4 - 1 = 3$ ✓ True.
2. **Test EQ 2** with x = 4, y = 3:
   $3 = 2(4) - 5 = 3$ ✓ True.
3. Both true, so **(4, 3) *is* a solution**. Even though it looked suspicious, the test confirms it.



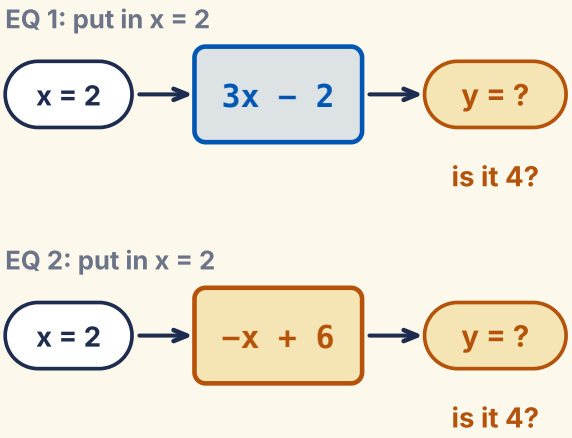
TAKEAWAY

You cannot tell by eyeballing. Always test **both** equations.

▶ Try It · Test the Point

> **EQ 1** $y = 3x - 2$
>
> **EQ 2** $y = -x + 6$

Is **(2, 4)** a solution to this system?



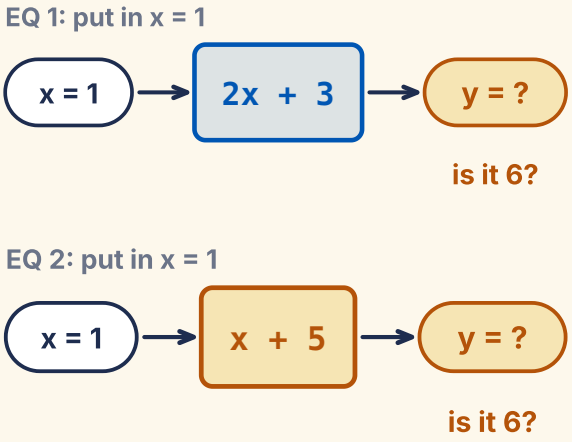
Circle one: YES NO

▶ Try It · Test the Point (2)

EQ 1 $y = 2x + 3$

EQ 2 $y = x + 5$

Is **(1, 6)** a solution to this system?

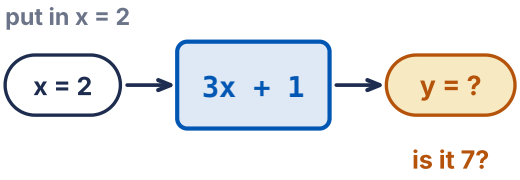


Circle one: **YES** **NO**

✏ Practice: What "Solution" Means

1. ★ Try it Does the point **(2, 7)** make **y = 3x + 1** true?

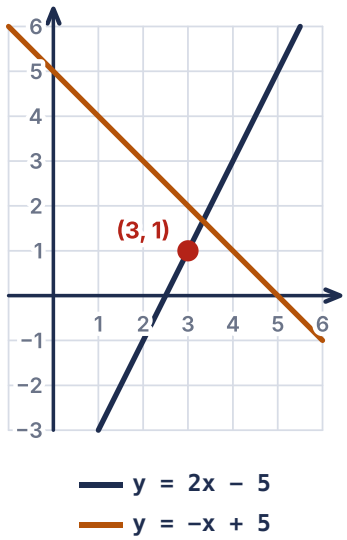
- ○ Yes
- ○ No



2. ★★ On level Is **(3, 1)** a solution of this system?

> **EQ 1** $y = 2x - 5$
>
> **EQ 2** $y = -x + 5$

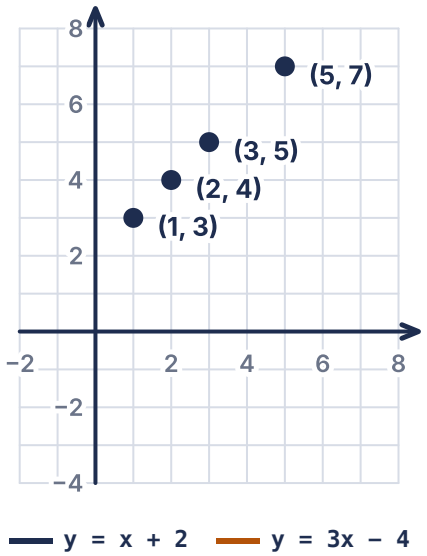
- ○ Yes
- ○ No



3. ★★★ Challenge Which point is the solution of this system?

> **EQ 1** $y = x + 2$
>
> **EQ 2** $y = 3x - 4$

- ○ (1, 3)
- ○ (2, 4)
- ○ (3, 5)
- ○ (5, 7)

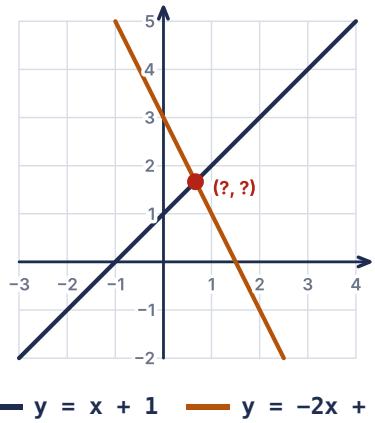


The 4 choices are plotted. Graph both lines to check.

4 Solving Systems by Substitution

Why Substitution?

Graphing is hard when the lines cross between grid lines. **Substitution** gives the exact answer.



These lines cross *between* grid lines. Reading the exact point is hard, so we use algebra.

The Steps

SUBSTITUTION PROCEDURE

1. Make sure one equation is solved for a variable (usually y = ...).
2. **Substitute** the expression into the *other* equation.
3. Solve that equation for x.
4. Plug x back in to find y.
5. Write the solution as an ordered pair (x, y) and check.

EXAMPLE 1 **Both equations in y = form**

EQ 1 $y = 2x + 3$

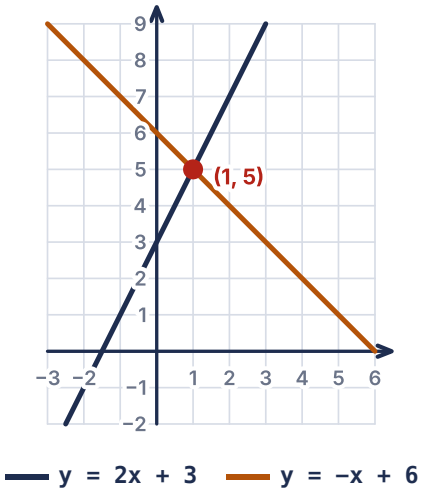
EQ 2 $y = -x + 6$

1. Both equations are solved for y. Since they both equal y, they must equal each other. Set them equal: $2x + 3 = -x + 6$
2. Solve for x. Add x to both sides: $3x + 3 = 6$. Subtract 3: $3x = 3$. Divide: $x = 1$.
3. Put your x into either equation to find y. Using EQ 1: $y = 2(1) + 3 = 5$.
4. **Solution: (1, 5)**
5. Check in EQ 2: $5 = -(1) + 6 = 5$ ✓

EQ 1 $y = 2x + 3$

EQ 2 $y = -x + 6$

so $2x + 3 = -x + 6$



EXAMPLE 2 **One equation in y = form, one not (Challenge)**

> **EQ 1** $y = 3x - 1$
>
> **EQ 2** $2x + y = 9$

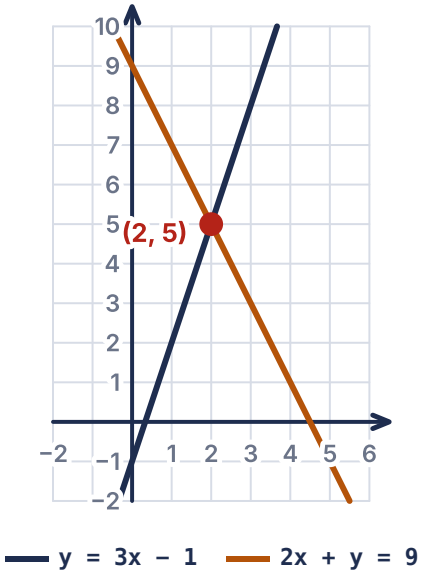
1. EQ 1 is already solved for y. So $y = 3x - 1$.
2. Substitute $(3x - 1)$ in place of y in EQ 2: $2x + (3x - 1) = 9$
3. Solve for x. Combine like terms: $5x - 1 = 9$. Add 1: $5x = 10$. Divide: $x = 2$.
4. Put your x into EQ 1 to find y: $y = 3(2) - 1 = 5$.
5. **Solution: (2, 5)**
6. Check EQ 2: $2(2) + 5 = 9$ ✓

EQ 1 $y = 3x - 1$

swap in

EQ 2 $2x + y = 9$

so $2x + (3x - 1) = 9$



EXAMPLE 3 **Quick rearrange required (Challenge)**

> **EQ 1** $x + y = 7$
>
> **EQ 2** $y = 2x + 1$

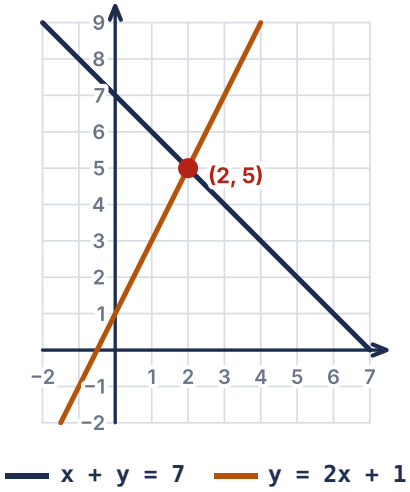
1. EQ 2 is already solved for y.
2. Substitute $(2x + 1)$ for y in EQ 1: $x + (2x + 1) = 7$
3. Solve: $3x + 1 = 7$ → $3x = 6$ → $x = 2$.
4. Put your x into EQ 2 to find y: $y = 2(2) + 1 = 5$.
5. **Solution: (2, 5)**

EQ 2 $y = 2x + 1$

swap in

EQ 1 $x + y = 7$

so $x + (2x + 1) = 7$



CHALLENGE NOTE

If *neither* equation is solved for a variable, pick the easier one and solve it first. If you see a coefficient of 1 on x or y, that equation is easiest.

▶ Try It · Substitution Practice 1

EQ 1 $y = x + 4$

EQ 2 $y = 3x - 2$

Solve the system using substitution. Write the solution as (x, y).

EQ 1 $y = \boxed{x + 4}$

EQ 2 $y = \boxed{3x - 2}$

so $\boxed{x + 4} = \boxed{3x - 2}$

Solution: (_____ , _____)

▶ Try It · Substitution Practice 2 ★★★ Challenge

EQ 1 $y = 2x - 3$

EQ 2 $x + y = 9$

Solve using substitution. (EQ 1 is already done for you — sub into EQ 2.)

EQ 1 $y = \boxed{2x - 3}$

swap in

EQ 2 $x + \boxed{y} = 9$

so $x + \boxed{(2x - 3)} = 9$

Solution: (_____ , _____)

▶ Try It · Substitution Practice 3 (quick rearrange) ★★★ Challenge

EQ 1 $2x + y = 10$

EQ 2 $y = x + 1$

EQ 2 is solved for y — sub (x + 1) for y into EQ 1.

EQ 2 $\quad y = \boxed{x + 1}$

swap in

EQ 1 $\quad 2x + \boxed{y} = 10$

so $\quad 2x + \boxed{(x + 1)} = 10$

Solution: (_____ , _____)

Graphing vs. Substitution — When to Use Which

GRAPHING

Good for seeing the answer. Hard when the answer is not a whole number.

SUBSTITUTION

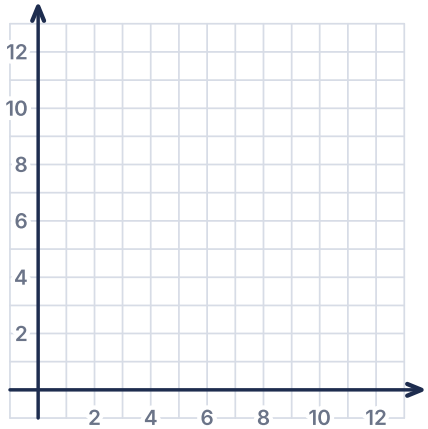
Best when at least one equation is already solved for x or y (or is easy to solve). Always gives an exact answer — fractions and all.

✏ Practice: Solving Systems by Substitution

1. ★★★ Challenge Solve for x. Write just the x-value.

> **EQ 1** $y = 2x$
>
> **EQ 2** $x + y = 12$

Answer: _____

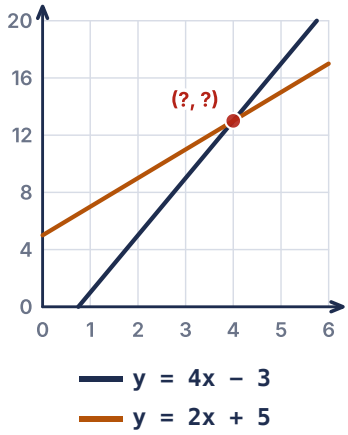


Check your answer: graph both lines here.

2. ★★ On level Solve by substitution. Write the ordered pair (x, y).

> **EQ 1** $y = 4x - 3$
>
> **EQ 2** $y = 2x + 5$

Answer: _____



The lines cross between grid lines. Substitution gives the exact point.

3. ★★★ Challenge Gym A costs y = 5x + 10 dollars and Gym B costs y = 2x + 25 dollars for x classes. After how many classes do they cost the same?

You can also finish the table. Look for the row where the costs match.

x	Gym A ($)	Gym B ($)
0	10	25
1	15	27
2	20	29
3		
4		
5		
6		

Answer: _____

5 Comprehensive Practice — All Sections

STANDARDS

NC.8.EE.8 — solve a system of two linear equations in slope-intercept form, and understand that the solution is the point where the graphs intersect. NC.8.F.4 — write a linear equation from a table. Parts A–C practice graphing; Part D mixes every section.

A. Read the Graph

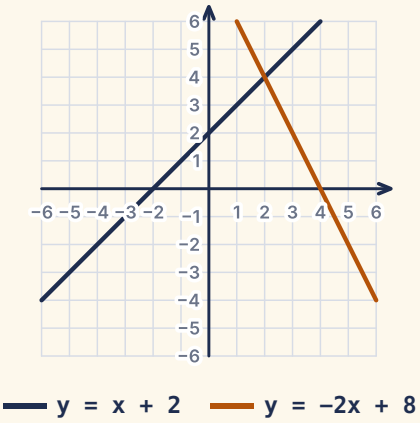
Both lines are already drawn. Find the point where they cross, then write that ordered pair.

1 · Read the intersection ★ Try it

EQ 1 $y = x + 2$

EQ 2 $y = -2x + 8$

Solution: (_____ , _____)



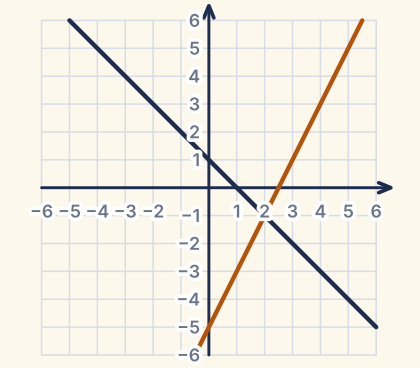
y = x + 2 y = −2x + 8

2 · Read the intersection ★★ On level

EQ 1 $y = -x + 1$

EQ 2 $y = 2x - 5$

Solution: (_____ , _____)



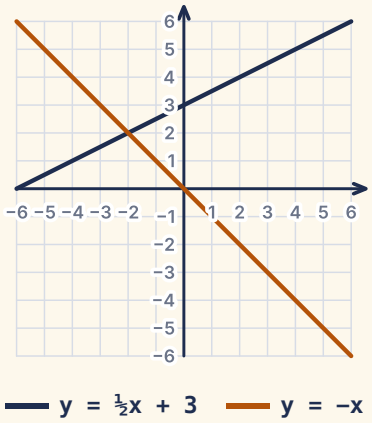
y = −x + 1 y = 2x − 5

3 · Read the intersection ★★★ Challenge

EQ 1 $y = \frac{1}{2}x + 3$

EQ 2 $y = -x$

Solution: (_____ , _____)



B. Graph It Yourself

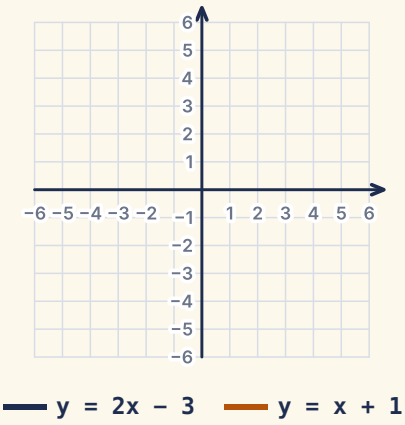
The grid is empty on purpose. Graph each line from its slope and y-intercept, read the intersection, then write the point where they cross.

1 · Graph and solve ★★ On level

EQ 1 $y = 2x - 3$

EQ 2 $y = x + 1$

Solution: (_____ , _____)

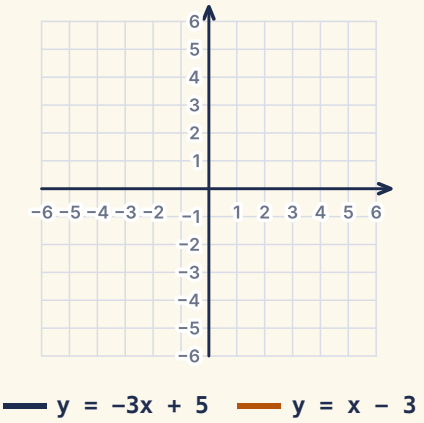


2 · Graph and solve ★★ On level

EQ 1 $y = -3x + 5$

EQ 2 $y = x - 3$

Solution: (_____ , _____)

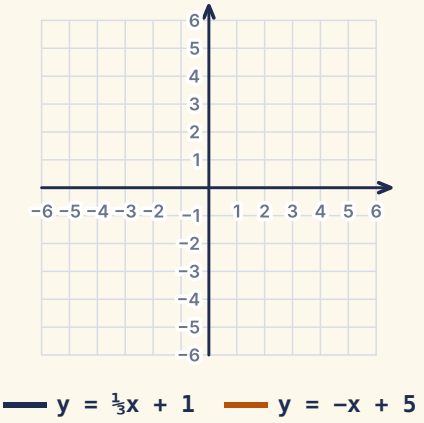


3 · Graph and solve ★★★ Challenge

EQ 1 $y = \frac{1}{3}x + 1$

EQ 2 $y = -x + 5$

Solution: (_____ , _____)



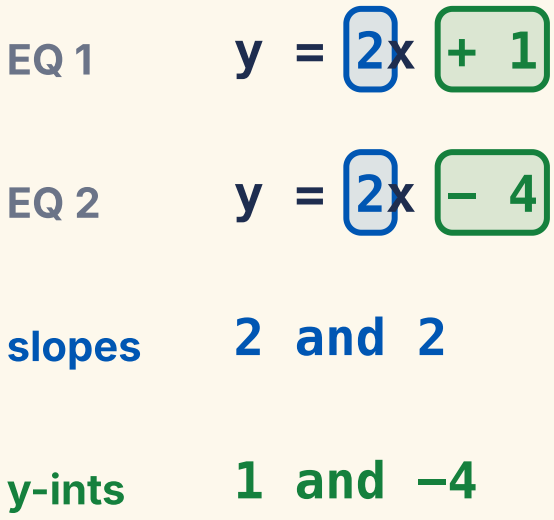
C. One, None, or Infinitely Many?

Compare the slopes and y-intercepts *before* you look at the picture.

1 · How many solutions? ★ Try it

EQ 1 $y = 2x + 1$

EQ 2 $y = 2x - 4$



Same or different? Decide.

Answer: _____

2 · How many solutions? ★★ On level

EQ 1 $y = 3x - 2$

EQ 2 $y = -x + 2$

EQ 1 $y = \boxed{3}x \boxed{- 2}$

EQ 2 $y = \boxed{-1}x \boxed{+ 2}$

slopes 3 and −1

y-ints −2 and 2

Same or different? Decide.

Answer: _____

3 · How many solutions? ★★ On level

EQ 1 $y = -\frac{1}{2}x + 3$

EQ 2 $y = -\frac{1}{2}x + 3$

EQ 1 $y = \boxed{-\frac{1}{2}}x \boxed{+ 3}$

EQ 2 $y = \boxed{-\frac{1}{2}}x \boxed{+ 3}$

slopes $-\frac{1}{2}$ and $-\frac{1}{2}$

y-ints 3 and 3

Same or different? Decide.

Answer: _____

D. Mixed Review — All Sections

✏ Comprehensive Practice Tables, graphing, solutions, substitution

1. ★ Try it In **y = 5x − 2**, what is b, the y-intercept? Write just the number.

Answer: _____

form $y = mx + b$

yours $y = 5x - 2$

Line them up. What is in the b spot?

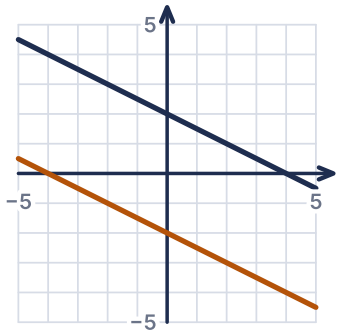
2. ★★ On level Write the equation for this table in the form y = mx + b.

Answer: _____

x	y
0	1
2	7
4	13
6	19

3. ★ Try it Two lines are parallel. How many solutions does the system have?

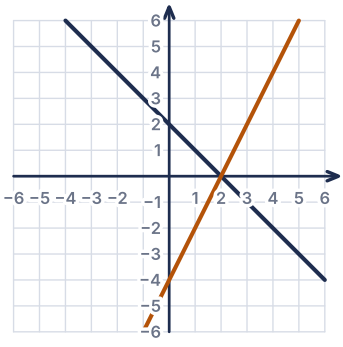
- ○ One solution
- ○ No solution
- ○ Infinitely many solutions



Two parallel lines

4. ★★ On level Where do the two lines cross? Write the ordered pair.

Answer: _____



y = −x + 2

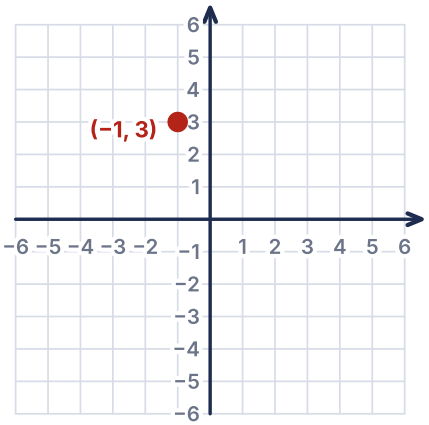
y = 2x − 4

5. ★★ On level Is **(−1, 3)** a solution of this system?

EQ 1 $y = -2x + 1$

EQ 2 $y = x + 4$

- ○ Yes
- ○ No

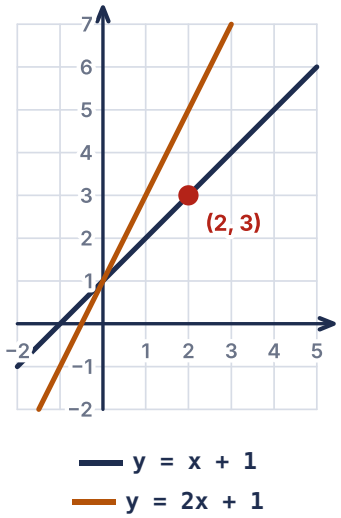


y = −2x + 1 y = x + 4

Graph both lines. Do both go through the point?

6. ★★★ Challenge Kai says (2, 3) is a solution of y = x + 1 and y = 2x + 1, because it works in the first equation. Is Kai right?

- ○ Yes, one true equation is enough
- ○ No, it must make both equations true



7. ★★★ Challenge Solve for x. Write just the x-value.

> **EQ 1** $y = 3x$
>
> **EQ 2** $x + y = 8$

Answer: _____

EQ 1 $y = 3x$

swap in

EQ 2 $x + y = 8$

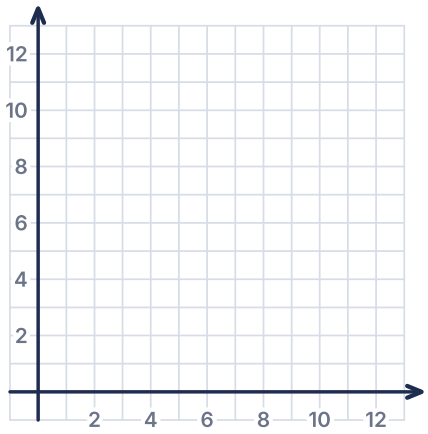
so $x + (3x) = 8$

8. ★★ On level Solve by substitution. Write the ordered pair (x, y).

EQ 1 $y = x + 6$

EQ 2 $y = 3x - 2$

Answer: _____



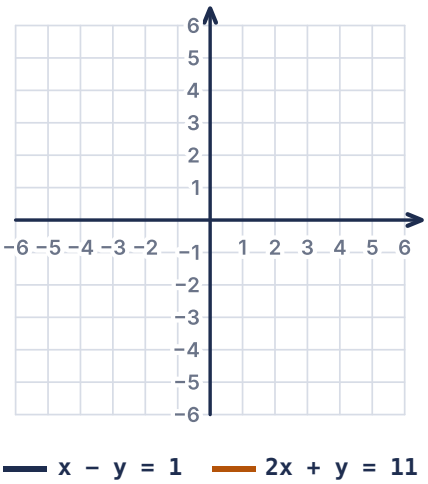
Check your answer: graph both lines here.

9. ★★★ Challenge Neither equation is solved for y. Solve the system and write the ordered pair.

EQ 1 $x - y = 1$

EQ 2 $2x + y = 11$

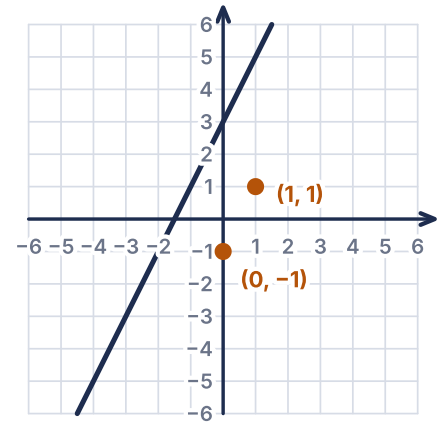
Answer: _____



Check your answer: graph both lines here.

10. ★★★ Challenge Line A is y = 2x + 3. Line B goes through (0, −1) and (1, 1). How many solutions does the system have?

- ○ One solution
- ○ No solution
- ○ Infinitely many solutions



Line A: y = 2x + 3

Line B: through the 2 points

Draw Line B through the two gold points.